Unit 6, Lesson 20

Interpret Remainders in Division Situations

Standards

Building On **4.OA.B.4**

Addressing **4.NBT.B.6, 4.OA.A.3**

Instructional Routines

- Choral Count

Goals

- Explain (orally and in writing) strategies for solving problems that involve interpreting a remainder.
- Interpret (orally and in writing) the quotient and remainder for a situation involving division.

Student Facing Learning Goals

Let's solve problems involving division and interpret remainders.

Lesson Purpose

The purpose of this lesson is for students to represent and solve contextual problems that involve dividing a whole number of up to four-digits by a single-digit divisor, resulting in a number with or without a remainder. Students also interpret the result and remainder given a situation.

Narrative

By now students have developed various strategies to divide multi-digit numbers by single-digit divisors and have used different representations along the way. In this lesson, students apply what they learned to solve a variety of word problems that involve division (MP2).

Access for Students with Disabilities

- Action and Expression

Access for English Learners

- MLR8

Lesson Timeline

Warm-up	10 min
Activity 1	15 min
Activity 2	20 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

What productive and unproductive beliefs did students show when they were solving problems today? How might you amplify the productive beliefs and address the unproductive ones?

Warm-up

10 min

Choral Count: 2, 3, and 5

Standards

Building On **4.OA.B.4**

Addressing **4.NBT.B.6**

Instructional Routines

- Choral Count

The purpose of this *Warm-up* is to invite students to count by 2, 3, and 5, notice patterns in the count, and think about multiples. It gives the teacher an opportunity to observe strategies or understandings students have for identifying multiples of small numbers, primarily by looking for and making use of structure (MP7). These understandings will be helpful later when students solve division problems that involve distinguishing quotients with and without a remainder.

Student Response

Counts by 2:

- 90 92 94 96
- 98 100 102 104
- 106 108 110 112

Counts by 3:

- 90 93 96
- 99 102 105
- 108 111 114

Counts by 5:

- 90 95 100
- 105 110 115

Launch

- "Count by 2 starting at 90."
- Record as students count.
- Stop counting and recording at 112.

Activity

- "What patterns do you notice in the recorded count?" (Count by 2: Ones digit is 0, 2, 4, 6, or 8.)
- Repeat with 3 and 5.
- "Count by 3 starting at 90."
- Stop counting and recording at 114.
- "What patterns do you notice in the recorded counts?" (Count by 2 and 3: The numbers 90, 96, 102, 108, and 114 are in both lists. These numbers are multiples of both 2 and 3. They have an add 6 pattern to get from one term to the next: $90 + 6 = 96$.)
- "Count by 5 starting at 90."
- Stop counting and recording at 115.
- "What patterns do you notice in the recorded counts?" (Count by 2, 3, and 5: The only number in all three lists is 90. 90 is a multiple of 2, 3, and 5. Count by 3 and 5: 90 and 105 are in both lists $90 + 15 = 105$.)

Activity Synthesis

- "Is 105 a multiple of 2, 3, or 5? How do you know?" (3 and 5. I know 90 is a multiple of 2, 3, and 5 so I can use the recorded counts to identify multiples. 105 is

in the recorded counts for 3 and 5.)

- "Is 105 a multiple of 15?" (Yes. I can count by 15s to get to 90: 15, 30, 45, 60, 75, 90 and then 15 more is 105.)

Activity 1

15 min

Hearts and Seats

Standards

Addressing **4.NBT.B.6, 4.OA.A.3**

The purpose of this activity is to encourage students to interpret the quantities in situations, represent them mathematically, use their representations to find solutions, and then interpret their solutions in context (MP2). The dividends here are limited to three-digit numbers.

Access for Students with Disabilities

Action and Expression: Develop Expression and Communication. Develop fluency with multiplication by 4 and 9. Provide a partially completed two-column table for each factor, and suggest that students record times tables that might be helpful for this activity before getting started. For example, in the left column of one table, students should complete a list of equations showing 9 times 1–10 ($9 \times 1 = 9, 9 \times 2 = 18 \ldots 9 \times 10 = 90$). In the right column, students should complete a list of equations showing 9 times multiples of 10 ($9 \times 10 = 90, 9 \times 20 = 180 \ldots 9 \times 100 = 900$). Repeat for 4, and invite students to use this reference they've made to complete the task. *Supports accessibility for: Conceptual Processing, Memory*

Student Task Statement

1. Two artists paint 378 wooden hearts. They sell the hearts in boxes of 4.
   - The first artist says they will need 94 boxes for all the hearts.
   - The second artist says 95 boxes are needed.

   Who do you agree with? Explain or show your reasoning.



2. An auditorium seats 258 people. The seats are arranged in rows of 9, but there is one short row with fewer than 9 seats.

   How many rows of 9 seats are there? How many seats are in the shorter row?

Launch

- Groups of 2

Activity

- 5–6 minutes: independent work time
- 2–3 minutes: partner discussion
- Monitor for students who appropriately interpret remainders based on the situation.

Activity Synthesis

- Invite students to share their responses and reasoning.
- For the first problem, highlight that both 94 and 95 boxes are plausible if they could be defended and the assumptions are made clear. (For example, students might say that the artists could have two

Student Response

1. Sample responses:
   - I agree with the first artist. $378 \div 4$ is 94 with a remainder of 2. The artists can't put the 2 extra hearts in a box and sell half a box, so they only need 94 boxes.
   - I agree with the second artist. If all the hearts are to be put in a box, there will be 94 boxes of 4 and 1 box with only 2 hearts. This means they will need 95 boxes.
2. There are 28 rows of 9 and 1 row of 6. Sample reasoning: 258 divided by 9 is 28 with a remainder of 6.

leftover hearts rather than trying to sell them in boxes, so 94 boxes are enough.)

- "What equation could we write to describe the relationship between the number of hearts and the number of full boxes?" (One possible equation: $(94 \times 4) + 2 = 378$.)
- For the last problem, ask: "What equation could we write to describe the relationship between the number of rows and the number of seats?" (One possible equation: $(28 \times 9) + 6 = 258$.)

Activity 2

20 min

Save for a Garden

Standards

Addressing **4.NBT.B.6, 4.OA.A.3**

The purpose of this activity is for students to continue to solve contextual problems that involve division (MP2). In this activity, the dividends extend to four-digit numbers and the problems demand a greater lift.

In the second half of the activity, students are asked to reason in the opposite direction: given a division expression, they are to invent a situation that can represent and interpret the value of the expression in context.

If time permits, consider asking students to create a visual display of the situation they invent for the second problem, so they can present the situation and their reasoning to the class.

Access for English Language Learners

MLR8 Discussion Supports. Synthesis: At the appropriate time, give groups 2–3 minutes to plan what they will say when they present to other groups. "Practice what you will say when you share your strategy with another group. Talk about what is important to say, and decide who will share each part."
Advances: Speaking, Conversing, Representing

Student Task Statement

1. A school needs $1,270 to build a garden. After saving the same amount each month for 8 months, the school still needs $6.

   How much did they save each month? Explain or show your reasoning.



Launch

- Groups of 2

Activity

- 3–4 minutes: independent work time on the first problem

2. Choose one of the following division expressions.

   $711 \div 3$ $3{,}128 \div 8$

   a. Write a situation to represent the expression.

   b. Find the value of the quotient. Explain or show your reasoning.

   c. What does the value of the quotient represent in your situation?

Student Response

1. $158 a month. Sample response:

   a. $1{,}270 \div 8 = 158$ with a remainder of 6. This means the school has saved up $1,264.

   b. $1{,}270 - 6 = 1{,}264$ and $1{,}264 \div 8 = 158$.

2. Sample response for $711 \div 3$:

   a. Three siblings collected 711 pennies and are dividing them equally. The quotient is the number of pennies each sibling gets.

   b. $711 \div 3 = 237$

   c. Each sibling gets 237 pennies.

   Sample response for $3{,}128 \div 8$:

   a. A family is making a road trip that is 3,128 miles long over 8 days. The quotient is the distance traveled each day if they travel the same distance per day.

   b. $3{,}128 \div 8 = 391$

   c. The family travels 391 miles a day.

- Invite students to share responses and reasoning.
- "What equation(s) can we write to represent the relationship between the amount saved each month, the savings, the number of months of saving, and the amount needed for the garden?" (One possible equation: $(8 \times 158) + 6 = 1{,}270$)
- 5–7 minutes: independent work time on second problem

Activity Synthesis

- Students find a partner who chose a different expression and they take turns presenting their work. The person listening should consider whether the response makes sense and check if the quotient is correct.
- If time permits, students share with a different student or partnership.

Lesson Synthesis

"Today we solved problems that involved division with and without remainders. What strategies did you find yourself using to divide numbers? Did you:

- use partial products?
- use partial quotients?
- draw diagrams?
- divide by place value (thousands, hundreds, tens, and ones)?
- write a series of equations?
- estimate first?"

"How do we know that there will be reminders when we divide?" (We know when the dividend is less than the divisor that we do not have enough to make another group or to put an equal amount in each group.)

"What if the remainder was four flowers left over after all the flower pots were filled, what would we do then?" (We could put more flowers in some pots or just give the extra flowers away.)

Suggested Centers

- Watch Your Remainder (4–5), Stage 1: One-Digit Divisors (Addressing)
- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)

Cool-down

5 min

Miscounting?

Standards

Addressing **4.NBT.B.6, 4.OA.A.3**

Student Task Statement

Mai is reciting multiples of 6. The last number she calls out is 194. Clare says, "I think you may have made a mistake."

Do you agree with Clare? Explain or show your reasoning.

Student Response

Yes, I agree with Clare. Sample responses:

- 194 is not a multiple of 6. I know that $6 \times 30 = 180$, and 194 is 14 away from 180. Because 14 is not a multiple of 6, 194 is also not a multiple of 6.
- Six is not a factor of 194. I divided 194 by 6 and got 32 with a remainder of 2. If Mai counted correctly, she would have called out 192 and then 198.

Responding to Student Thinking

Students do not agree with Mai and justify it with faulty reasoning. (For example, they may think that 200 is a multiple of 6. Therefore, 194, which is 6 less than 200, is also a multiple of 6.)

Next Day Supports
Before the *Warm-up*, invite students to work in small groups to discuss a correct response to this *Cool-down*.